Observability

English version of ../../operations/monitoring.md.

Signal	Where	Purpose
/healthz	backend	the process is alive
/readyz	backend	readiness, including database availability
Security headers	backend/middleware/security_headers.py	checked by tests/test_security_headers.py
Action audit	backend/middleware/audit.py, backend/api/audit.py	who changed what
Rate limit	backend/middleware/rate_limit.py	brute-force protection
Report cache	Redis/KeyDB (services/report_cache.py)	speeds up aggregates; when unavailable, computed directly
Notifications	services/scheduler.py	deadline reminders

Minimal alerts

1. /readyz does not answer for 2 minutes (unlike /healthz, this means a dependency failure).
2. The 5xx share is above 1% over 5 minutes.
3. Redis is unavailable — reports slow down; this is degradation, not an outage.
4. GigaChat errors while summarisation is enabled.
5. A growing number of 429s (rate limit) — possible brute force or a load spike.

What is missing

- No /metrics and no ready-made dashboards: observability rests on logs, health checks and audit.
- No distributed tracing.
- No SLOs expressed as numbers.